



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (VDE) (CBCS) Degree Examinations

May – June 2025

22VDE233 – SCRIPTING LANGUAGES FOR VLSI

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Write a program that prompts for and reads two numbers, and prints out the result of the two numbers multiplied together.	8	L*3	1	3
b) Explain auto increment and auto decrement in PERL	6	L2	2	3
c) Explain the chop and chomp functions with an example.	6	L2	2	3
2. a) Write a program that reads a number and then a list of strings (all on separate lines), and then prints one of the lines from the list as selected by the number.	8	L3	1	3
b) Explain Hash Functions.	6	L2	2	3
c) Explain the while/until Statement.	6	L2	2	3
3. a) Demonstrate with an example, VHDL Test bench creation using Perl.	8	L3	1	3
b) Write a procedure for test vectors generation in Perl.	6	L2	2	3
c) Explain the Working of Netlist Patching.	6	L2	2	3

Unit – II

4. a) Write the rules that define the syntax and semantics of the Tcl language (any eight rules).	8	L2	3	3
b) Write the format and example for the following string commands String compare, String index, String length, String to lower, String to upper, String range	6	L2	4	3
c) Write a program in tcl that uses the switch command.	6	L3	4	3
5. a) Write a program to create a procedure which transforms Fahrenheit temperature to Celsius temperature.	8	L3	3	3
b) Explain the while command.	6	L2	4	3
c) Explain the switch command.	6	L2	4	3
6. a) Explain the procedure in tcl for single, double and for a variable number of arguments.	8	L2	3	3
b) Write a note on Tk Widget.	6	L2	4	3
c) Make use of Tcl/Tk to develop graphical program to display Hello World!.	6	L3	4	3

Unit – III

7. a) Write the Fundamental Verification Principle and write the procedure for Specification Design Decomposition.	10	L2	5	3
b) Write the Principle behind the Retain Useful Information, Explain the High-Level Design Requirements with a neat diagram.	10	L2	5	3
8. a) What is Random Test. With a neat diagram, explain hybrid random simulation environment.	10	L2	5	3
b) How to make sure the Functional Correctness, explain the Programming Code Metrics.	10	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
